nth Roots, Radicals, and Rational Exponents

Algebra 2 Lesson 5-1 Name _____ Date _____ Period _____

Learning goal: I can relate rational exponents to roots, evaluate expressions with rational exponents, and simplify nth roots of expressions with variables.

Language goal: I can explain what a rational exponent means, and how I simplified an nth root, using the words root, power, index and radicand.

Spanish: *índice* (index), *radicando* (radicand), *exponente* (exponent), *símbolo radical* (radical symbol).

On the quiz: (1) real fourth roots (full Savvas quiz only; Example 1). (2) pick every form equal to a rational exponent and (3) explain one, then evaluate it: Example 2, Try It 2, Practice. (4) simplify an nth root with variables: Example 4, Try It 4. (5) solve $ax^3 = c$ (full Savvas quiz only; Example 5). Questions 3 and 4 count 10 points each; 2 is the +5 bonus.

On the Topic 5 test: Question 1 asks you to simplify $\sqrt[3]{27x^9y^3}$ (Example 4). Questions 3, 5 and 10 ask you to match a rational exponent with its radical; Question 18 asks which property defines $x^{\frac{a}{b}}$ (Example 2).

Example 2: Understand Rational Exponents

A. What is the meaning of the exponent in the expression $16^{\frac{1}{4}}$?

Extend the properties of integer exponents.

$$16^{\frac{1}{4}} = \left(2^4\right)^{\frac{1}{4}} \qquad 16 = 2^4$$
$$= 2^{4\cdot\frac{1}{4}} = 2^1 \qquad \text{Power of a Power}$$
$$= 2$$

Since $2^4 = 16$, $16^{\frac{1}{4}}$ is $\sqrt[4]{16}$, the fourth **root** of 16, which is 2.

In general, $x^{\frac{1}{n}} = \sqrt[n]{x}$, the principal nth root.

B. Write $27^{\frac{2}{3}}$ as a radical.

$$27^{\frac{2}{3}} = \left(27^{\frac{1}{3}}\right)^2 \qquad \text{Power of a Power}$$
$$= \left(\sqrt[3]{27}\right)^2$$
$$= 3^2 = 9$$

27 is a perfect cube (3^3), so take the cube root first, then square: $\left(\sqrt[3]{27}\right)^2 = 9$.

Same value the other way (power first, then root):

$\sqrt[3]{27^2} = \sqrt[3]{729} = 9$.

The rule. When $\sqrt[n]{c}$ is real, $c^{\frac{m}{n}} = \left(\sqrt[n]{c}\right)^m = \sqrt[n]{c^m}$.

Denominator n = the **index** (which **root**); numerator m = the **power**; c = the **radicand**.

Say it: In $27^{\frac{2}{3}}$, the denominator 3 is the _____ of the radical, so I take the cube _____. The numerator 2 is the _____, so I square. The _____ is 27.

Try It 2: Explain what each fractional exponent means, then evaluate

a. $25^{\frac{1}{2}}$

$25^{\frac{1}{2}}$ is the _____ root of 25.

_____

_____

Value: _____

b. $32^{\frac{2}{5}}$

$32^{\frac{2}{5}}$ is the _____ root of 32, _____.

_____

_____

Value: _____

Practice: Which forms are equal?

Select **all** the expressions that are equivalent to $y^{\frac{2}{5}}$. Use the rule: denominator = index, numerator = power.

☐ **A.** $\sqrt[5]{y^2}$ ☐ **B.** $\left(y^{\frac{1}{5}}\right)^2$ ☐ **C.** $\sqrt{y^5}$

☐ **D.** $y^{\frac{10}{4}}$ ☐ **E.** $(y^2)^{\frac{1}{5}}$ ☐ **F.** $\left(y^{\frac{1}{2}}\right)^{\frac{1}{5}}$

For one choice you did *not* select, write it as a single power of y and say what went wrong.

Example 4: Simplify nth Roots

Simplify each expression. Assume all variables are positive.

A. $\sqrt[5]{32m^{15}}$

$$\begin{aligned}\sqrt[5]{32m^{15}} &= \sqrt[5]{2^5 m^{15}} && 32 = 2^5\\ &= \sqrt[5]{(2m^3)^5} && \text{radicand as a 5th power}\\ &= 2m^3\end{aligned}$$

Write the **radicand** as an expression to the 5th **power**; the fifth **root** of $(2m^3)^5$ is $2m^3$.
Check: $(2m^3)^5 = 2^5 m^{3\cdot 5} = 32m^{15}$.

B. $\sqrt[4]{x^{20}y^8}$

$$\begin{aligned}\sqrt[4]{x^{20}y^8} &= \left(x^{20}y^8\right)^{\frac{1}{4}} && \text{raise to the } \tfrac{1}{4}\\ &= x^{\frac{20}{4}}y^{\frac{8}{4}} && \text{Power of a Power}\\ &= x^5y^2\end{aligned}$$

Or use Part A's method: 20 and 8 are both multiples of the **index** 4, so $\sqrt[4]{x^{20}y^8} = \sqrt[4]{(x^5y^2)^4} = x^5y^2$.

The move. Write the radicand as an nth power and use $\sqrt[n]{a^n} = a$ (true for $a \geq 0$, and for every a when n is odd); or raise the radicand to the $\frac{1}{n}$ power and divide every exponent by the index n. Why "all variables positive": for an even index, the radical denotes the nonnegative principal root; the positive-variable assumption lets us write x^5y^2 without absolute values. With an odd index a negative radicand is fine: $\sqrt[3]{-8} = -2$.

Say it: In $\sqrt[4]{x^{20}y^8}$, the _____ is 4 and the _____ is $x^{20}y^8$. I wrote it as a fourth _____, $(x^5y^2)^4$, so the fourth _____ is x^5y^2.

Try It 4: Simplify each expression

a. $\sqrt[3]{-8a^3b^9}$

Answer: _____

b. $\sqrt[4]{256x^{12}y^{24}}$

Answer: _____

Say it: In part a, the _____ 3 is odd, so the cube _____ of a negative radicand is _____ (positive / negative).

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
9TM ★	Convert between rational exponents and radicals	Quiz 2, 3 (Example 2)
KJX ★	Evaluate rational exponents	Quiz 3 (Example 2, Try It 2)
LQX ★	Simplify radical expressions with variables I	Quiz 4 (Example 4, Try It 4)
U42	Nth roots	Quiz 1, 5 (full Savvas quiz only), optional
EUH	Roots of integers	Warm-up, optional: the perfect powers 32, 243, 81, 256

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Extension (self-paced): Practice 50 (DOK 3, Performance Task)

A milk processing company uses cylindrical-shaped containers. The height of the container is equal to the diameter of the base. The volume of a container is 169.65 ft^3.

Part A How much material is needed to make the lateral surface area of the shipping container?

Part B The cargo hold of a ship is 20 ft high. What is the largest number of these shipping containers that could be stacked on top of each other inside the cargo hold?

Hint: write V using r only; then r^3 is a number, and r is its cube root, as $\sqrt[3]{27} = 3$ in Example 2B.

Part A: _____ ft^2 Part B: _____ containers

Before the quiz, finish these sentences.

In $c^{\frac{m}{n}}$, the denominator n is the _____, the numerator m is the _____, and c is the _____.

To simplify the nth root of a product, I write the radicand as an nth _____, or I divide every exponent by the _____ n. With an even index the radicand must not be _____; with an odd index a negative radicand gives a negative _____.